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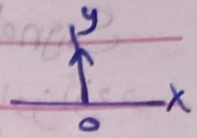
Digital waveforms GEneration

(A Periodic Signal) → Sinc func

Impulse ← unitstep ↓ Ramp ↓ → Rectangular Pulse → Triangle Pulse

* Impulse function = [---0000100000---]



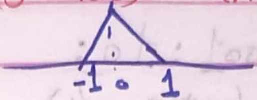
% A Periodic Signals

% Impulse function المفروض يبقى كده

nts = -10:10; (X-axis) nts=-10:0.001:10;

Signal = nts == 0; (y-axis) amplitude 1 at Time 0

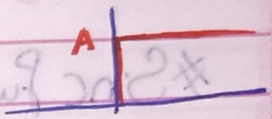
Plot (nts, Signal).



لازم أعمل step صغيرة

Signal = 3* (nts ==0); لو عاوز أزود ال Amplitude

*unit step function = [---000 111---]

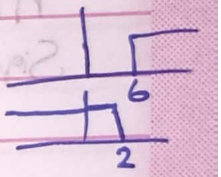


% unit step function

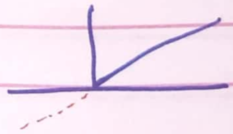
nts = -10:0.001:10;

Signal = nts >=0; Signal = nts >= 6; يظهر عند النقطة دي

Plot(nts, Signal) nts <= 2;



*Ramp function = [--000 1234---]



% Ramp function

nts = -10:0.001:10;

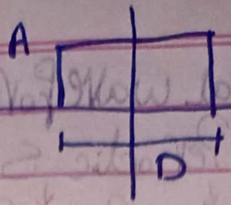
Signal = nts .*(nts >=0); ضربتها في unit step علشان ألغي الجزء السالب

Plot(nts, Signal)

(nts-4) .* (nts >= 4)

لو عاوز أشفتها عند ال ④

*Rectangular Pulse



```
% Rectangular Pulse
nts = -10:0.001:10;
Signal = 5* rectPuls(nts, 4);
```

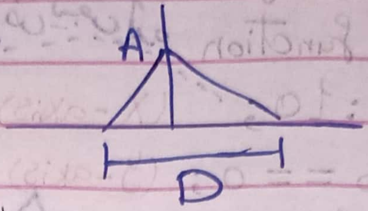
لو عاوز أزود Amplitude ← 5*

أمر مبتاع ال ← rectPuls

بداية ونهاية ← nts

Duration ← 4

*Triangle Pulse



```
% Triangle Pulse
nts = -10 : 0.001 : 10;
Signal = triPuls(nts, 4); (nts, 4, -1)
```

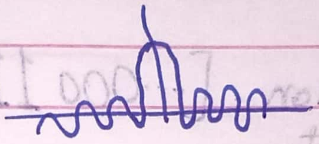
أمر مش ← triPuls

بداية ونهاية ← nts

Duration ← 4

قيمة ↳slope -1 : 1

*Sinc Function



```
% Sinc function
nts = -10:0.001:10;
Signal = Sinc(nts);
```

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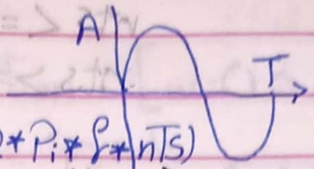
Digital wave form generation

(Periodic Signals)

Sine ← Sawtooth → Triangle → Square

Sine wave



Signal = A * Sin(2 * Pi * f * nTS)

Freq. ← f   nTS → x-axis بداية : خطوة : نهاية

Ex: generate Sine wave with freq 1KHz And AmPiTude 2V

%Sine wave   أحسب Sampling freq الأول

$f_s \geq 2 * f_{max}$

$f_s \geq 2 * 1000$

$f_s \geq 2000$ $\therefore F_s = 10000$

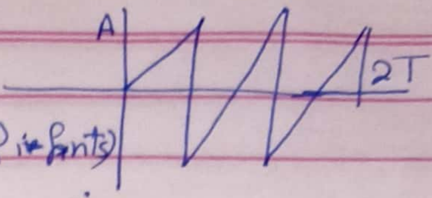
```
%Sine wave
  FS = 10000;
  TS = 1/FS;
  F = 1000;
  T = 1/F;
nts = 0:Ts:T;
Signal = Sin(2*Pi*F*nts);
Plot(nts, signal);
```

FS = 10000 ← أزوده عشان نظبط شكل الـ Signal

nts = 0:Ts:T; كده هشوف Cycle واحدة

لو عايز أكتر من واحدة أخلي $\boxed{3}$T → أي عدد Cycle أنا عايزه

Sawtooth wave



Signal= A*SawTooth(2*Pi*f*nts)

نفس طريقة الـ Sine

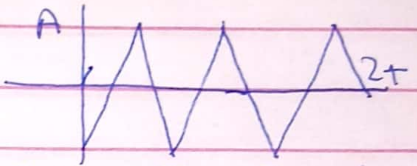
لازم أجيب (Fs)

```
%Sawtooth wave
Fs=10000;     Ts=1/Fs;     F=1000;     T=1/F;
nts=0:Ts:2T;
Signal=2*SawTooth(2*Pi*F*nTS);
Plot(nTS,Signal);
```

2T → 2 cycle هنشوف

2 → Amplitude

Triangle wave

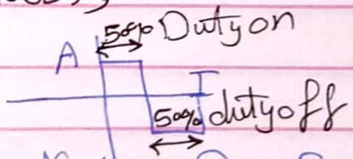


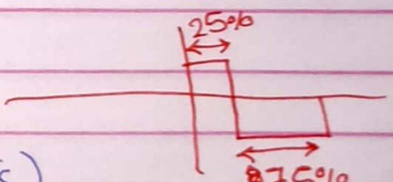
Signal= A*SawTooth(2*Pi*f*nts, 0.5)

Slope عشان يطلعلي الTriangle

```
% Triangle wave
Fs=10000;    TS=1/Fs;    F=1000;    T=1/F;
nTS=0:TS:T;
Signal=2*SawTooth(2*Pi*F*nts,0.5);
Plot(nTS,Signal);
```

Square wave





Signal= A*Square(2*Pi*f*nts)

Square(2*Pi*f*nts, Duty on)

```
%Square wave
Fs=100000;    TS=1/Fs;    F=1000;    T=1/F;
nTS=0:TS:T;
Signal=3*Square(2*Pi*F*nts);
Plot(nts,signal);
```

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FAST FouRieR TransfoRm

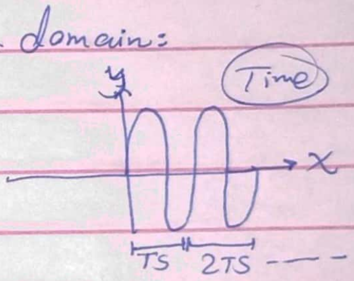
$$x(t) = 2\cos(1000\pi t) + 3\cos(2000\pi t) + 5\cos(4000\pi t)$$

$f_1 = 500$
$f_2 = 1000$
$f_3 = 2000$

SamPling freq using Nyquist :

$f_s \geq 2 * f_{max}$
$f_s \geq 2*2000$
$f_s \geq 4000$
$\boxed{f_s = 10000}$

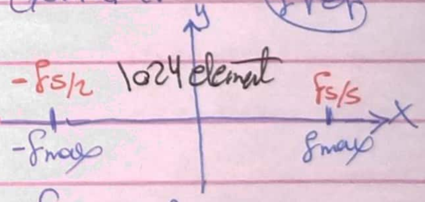
# Generat Signal in Time domain → Freq domain:



```
Fs=10000;
Ts=1/Fs;
N= 1024;     N.of SamPle
nts=0:Ts:(1023*Ts);
Signal= 2*Cos(1000*Pi*nts)+ 3*Cos(2000*Pi*nts)
        + 5*Cos(4000*Pi*nts);
```

# transform signal from Time To freq domain:

```
Signalf= fft(Signal);   بيحول من T→f
Signalff= fftShift(Signalf);   بيغير الـ center
F= lineSPace(-Fs/2, Fs/2, 1024);
                           (N)
Figure(1)
Plot(F, abs(Signalff));
Figure(2)
Plot(F, angle(Signalff))
```



$f_s \geq 2 * f_{max}$
$\therefore f_{max} \leq f_s/2$
$\therefore f_{max} = f_s/2$

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#generate Signal in Time domain:

```
N=1024; TS=(10--10)/N;  Fs=1/TS;
nts=-10:(0.01):10;
Signal=rectPuls(nts);
Figure(1)
Plot(nts,Signal)
```

nts=linsPace(-10,10,N);

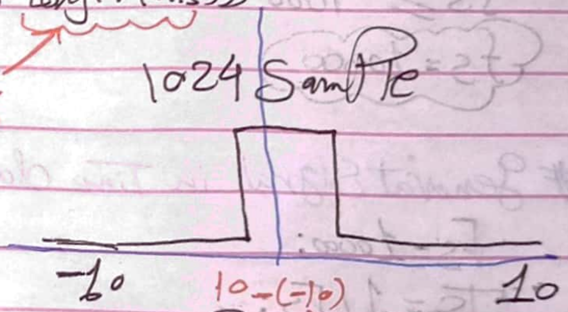
LAB1 ممكن ع تشتغل ع أي سيجنال

# transform signal into freq domain

```
Signalf=fft(Signal);
Signalff=fftShift(Signalf);
F=linsPace(-Fs/2,Fs/2,length(nts));
Figure(2)
Plot(F,abs(Signalff))
Figure(3)
Plot(F,angle(Signalff))
```

Signalff=fftshift(fft(Signal));

ممكن أحطها مكانها (N) =



$$TS = \frac{20}{1024} = \boxed{}$$

$$Fs = 1/TS$$

LAB 3

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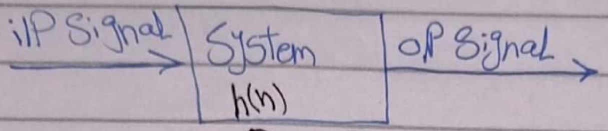
Convolution ↓

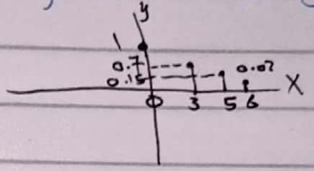


Impulse Response

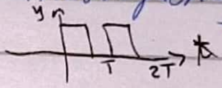
Ex: Estimate Signal at the Receiver assuming the Channel is Multi Path with:

(y). Coefficients = [1 0.7 0.15 0.02]

(x). Delay = [0 3 5 6]

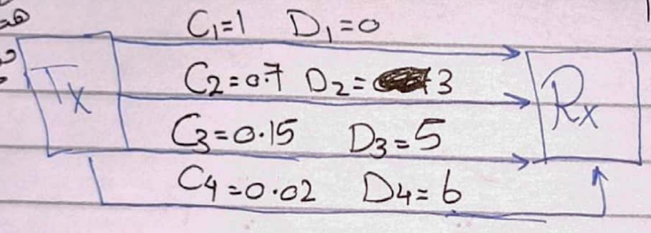


And the input Signal is a Square wave with Frequency 1000 Hz



※ → h(n) = [1 0 0 0.7 0 0.15 0.02]

عند التانية زيرو هحط ٢ عناصر وهكذا ...



$C_1=1$ $D_1=0$

$C_2=0.7$ $D_2=$ [redacted] 3

$C_3=0.15$ $D_3=5$

$C_4=0.02$ $D_4=6$

عدد ال element اللي موجوده في ال h(n)
← آخر Delay element
زود عليه واحد #6+1=7

$f_1 = 1000$ Hz → $f_s \geq 2 * f_{max}$ → $f_s \geq 2000$ Hz

1 %generating input Signal

```
Fs = 100000;
Ts = 1/Fs;
F = 1000;
T = 1/F;
nts = 0 : Ts : 2T;
Signal = Square (2*Pi*1000*nTS);
```

(x-axis) ← nts

اكرر كذا ال cycle

(F) ↓ 1000

3 % Estimation of output Signal

```
outSignal = Conv (Signal, hn)
Figure (1)
Plot (Signal)
Figure (2)
Plot (outSignal)
```

2 %Determination of impulse Response

```
Coefficients = [1 0.7 0.15 0.02]
Delay = [0 3 5 6]
hn = Zeros(1, 7)
hn(Delay+1) = Coefficients ⟹ h(n)
```

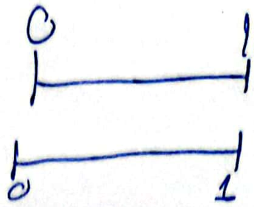
Zeros(1 ← Row, 7 ← column)

عشان أول element zero والماتلاب أول index هو واحد

عشان يطلعلي ال h(n)

مهم

بسيب ال distribution عشان ال Coefficients بتاعتي مختلفة لكل Path #



نقسم كل الأعداد على أكبر عدد

minimization

LAB4

DIGITAL FILTERS

$f_s \geq 2 * f_{max}$

$\left(f_{max} = \frac{f_s}{2}\right)$

Type of digital filters:

- FINITE Impulse Response (FIR)
- Infinite Impulse Response (IIR) ✓✓✓
  - Butterworth ✓✓✓
  - Chebyshev
  - Elliptic

T.F of Butterworth in Z-domain

$$H(z) = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} - -}{1 + b_1 z^{-1} + b_2 z^{-2}}$$

Using MATLAB Software design LPF with Following Specification:

Order = 10 Cut-off freq. = 600 HZ

where input signal is:

$x(t) = 2\cos(1000\pi t) + 3\cos(2000\pi t) + 5\cos(4000\pi t)$

and number of sample = 1024

```
%Generate input signal
 Fs=10000;
 Ts=1/Fs;
 N=1024;
nTS=0:Ts:(N-1)*Ts
Inputsignal=2*cos(1000*pi*nTs)+3*cos(2000*pi*nTs)
            +5*cos(4000*pi*nTs)
%Filtering Process
 order=10;
 Fc=600;
[a b]=butter(order,Fc/(Fs/2),'Low');
Outsignal=Filter(a,b,inputsignal);
```

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```
inPutSignalF=fftShift(fft(inPutsignal'));
outSignalF=fftShift(fft(outsignal));
F= linspace(-Fs/2 , Fs/2 ,N)
Figure(1)
Plot(F, abs(InPutsignalF))
Figure(2)
Plot( F, abs(outsignalF))

Figure(3)
Freqz(a, b, N, Fs)
```